

A fixed-character gap for abelian-by-extraspecial kernels in the commuting-probability $1/97$ problem

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

The existence of a finite group of odd order with commuting probability $1/97$ remains unresolved by this note. We exclude a specified infinite family from the necessary kernel equation in Beyer de Ryke's least-order reduction. Let E be an extraspecial 3-group and let N be a nontrivial abelian group of odd order prime to 3, on which E acts with image of order 3 and no nonidentity common fixed point. For $K = N \rtimes E$, no automorphism whose cube is inner satisfies that equation. A class count reduces the proof to $|N| = 7$ and $|E| = 243$, where the equation requires exactly 107 fixed irreducible characters. We prove instead that the fixed count is at most 95, or is 137 or 149. Both extraspecial exponents and nonsplit index-three extensions are included.

1 The kernel equation and the exclusion

For a finite group X , let $k(X) = |\text{Irr}(X)|$ be its number of conjugacy classes, where $\text{Irr}(X)$ denotes the irreducible complex characters of X . Its commuting probability is

$$\text{cp}(X) = \frac{|\{(x, y) \in X^2 : xy = yx\}|}{|X|^2} = \frac{k(X)}{|X|}.$$

Beyer de Ryke's work on Kourovka problem 21.88 leaves the odd-order existence question at $1/97$ unresolved in the cited version [1]. In its least-order reduction, a hypothetical witness has the form $P \rtimes H$, with P an abelian normal Sylow 97-subgroup and $H/K \cong C_3$ for $K = C_H(P)$ [1, Theorem 5.1]. In particular, writing $|P| = 97^e$, the required equation is

$$8|\text{Irr}(K)^H| = 97^{e-1}(9|K| - 97k(K)), \quad e \geq 1. \quad (1)$$

If $t \in H$ lifts a generator of H/K , conjugation by t induces $\alpha \in \text{Aut}(K)$ with $\alpha^3 \in \text{Inn}(K)$, and $\text{Irr}(K)^H = \text{Irr}(K)^\alpha$. Here a character is fixed when $\chi \circ \alpha = \chi$. Put $f(\alpha) = |\text{Irr}(K)^\alpha|$.

Recall that a 3-group E is extraspecial if $Z(E) = E' = \Phi(E)$ has order 3, where $\Phi(E)$ is its Frattini subgroup. Its order is 3^{2n+1} for some $n \geq 1$.

Theorem 1. *Let E be any extraspecial 3-group of order 3^{2n+1} , $n \geq 1$. Let N be a nontrivial finite abelian group of odd order prime to 3, and let $\rho : E \rightarrow \text{Aut}(N)$ satisfy*

$$|\rho(E)| = 3, \quad C_N(\rho(E)) = 1.$$

Set $K = N \rtimes_\rho E$, with N the normal factor. There are no integer $e \geq 1$ and $\alpha \in \text{Aut}(K)$ such that $\alpha^3 \in \text{Inn}(K)$ and

$$8f(\alpha) = 97^{e-1}(9|K| - 97k(K)). \quad (2)$$

Consequently no such K occurs as the kernel in the least-order reduction above.

The theorem covers extraspecial groups of exponent 3 and exponent 9 and every action satisfying its hypotheses. Its local input is the following gap.

Lemma 2. *Under the hypotheses of Theorem 1, suppose $|N| = 7$ and $|E| = 243$. If $\alpha^3 \in \text{Inn}(K)$, then*

$$f(\alpha) \leq 95, \quad \text{or} \quad f(\alpha) = 137, \quad \text{or} \quad f(\alpha) = 149.$$

The closest exclusion in [1, Corollary 5.9] concerns extraspecial kernels themselves. The groups here have a nontrivial normal abelian Hall $3'$ -subgroup acted on nontrivially by E , and are nonnilpotent. The additional step is control of the automorphism action on characters after the class-count inequalities have been satisfied. No equivalent exclusion was identified in the inspected source and limited literature searches; this is not an assertion of priority. For broad adjacent finite-group context, Zeng's work on equal-sign fiber products also studies group structure through normal subgroups and quotients [2]; it supplies no commuting-probability or fixed-character result used here.

2 Class counts and the remaining parameters

Write $s = |N|$, $Z = Z(E)$, $V = E/Z$ and $T = \ker \rho$. We regard V as a $2n$ -dimensional vector space over $\mathbb{F}_3$.

Lemma 3. *With this notation,*

$$k(E) = 3^{2n} + 2, \quad k(T) = 3^{2n-1} + 6, \quad (3)$$

$$k(K) = 3^{2n-2}(s+8) + 2s, \quad \text{cp}(K) = \frac{s+8}{27s} + \frac{2}{3^{2n+1}}. \quad (4)$$

Moreover, $s \equiv 1 \pmod{3}$, so $s = 7$ or $s \geq 13$.

Proof. Since $\rho(E)$ is abelian, $Z = E' \leq T$. Commutation induces a nondegenerate alternating bilinear form $V \times V \rightarrow Z \cong \mathbb{F}_3$. The space $W = T/Z$ is a hyperplane. Its perpendicular $W^\perp$ is a line contained in W , and hence is the radical of the restricted form. Therefore $|Z(T)| = 9$. A noncentral element of E has conjugacy class of size 3: its commutator map has nontrivial image in Z . The same argument applies to noncentral elements of T . Thus

$$k(E) = 3 + \frac{3^{2n+1} - 3}{3}, \quad k(T) = 9 + \frac{3^{2n} - 9}{3},$$

which gives (3). When $n = 1$, T is abelian of order 9, so this case is included.

Let σ generate $\rho(E)$. The map $x \mapsto x^{-1}\sigma(x)$ is an endomorphism of N with trivial kernel, hence is bijective. Consequently $[N, E] = N$. An E -invariant linear character of N is trivial on $[N, E]$, so only the principal character is invariant. Every nonprincipal character has an orbit of length 3 and inertia subgroup $N \times T$ in K . In particular, $3 \mid s - 1$. The conditions that $s > 1$ is odd give $s = 7$ or $s \geq 13$.

The irreducibles with N in their kernel are inflated from E . For each other orbit, fix a representative $\lambda \in \text{Irr}(N)$. The characters $\lambda \otimes \theta$, $\theta \in \text{Irr}(T)$, are exactly the irreducibles of $N \times T$ above λ . Their inductions to K are irreducible and are pairwise distinct, by Clifford correspondence. Equivalently, the three distinct conjugates of λ separate the summands on restriction to N ; an invariant subspace is then determined by its part above λ , on which θ is irreducible. These induced characters exhaust the irreducibles over the chosen orbit. Hence

$$k(K) = k(E) + \frac{s-1}{3}k(T) = 3^{2n-2}(s+8) + 2s.$$

Division by $|K| = s3^{2n+1}$ proves (4). □

Every automorphism fixes the principal character, so $1 \leq f(\alpha) \leq k(K)$. If (2) holds, positivity gives $\text{cp}(K) < 9/97$. Also,

$$9|K| - 97k(K) = \frac{8f(\alpha)}{97^{e-1}} \leq 8k(K),$$

so necessarily

$$\frac{3}{35} \leq \text{cp}(K) < \frac{9}{97}. \quad (5)$$

When $n = 1$, formula (4) gives $\text{cp}(K) = 1/9 + 8/(27s) > 9/97$. The same formula decreases in each of s and n . If $n \geq 2$ and $s \geq 13$, it gives

$$\text{cp}(K) \leq \frac{21}{27 \cdot 13} + \frac{2}{243} = \frac{215}{3159} < \frac{3}{35}.$$

If $n \geq 3$ and $s = 7$, it gives

$$\text{cp}(K) \leq \frac{15}{27 \cdot 7} + \frac{2}{2187} = \frac{1229}{15309} < \frac{3}{35}.$$

Thus the only remaining parameters are $s = 7$ and $n = 2$. They yield

$$|K| = 1701, \quad k(K) = 149, \quad 9|K| - 97k(K) = 856 = 8 \cdot 107. \quad (6)$$

Equation (2) now requires $f(\alpha) = 107 \cdot 97^{e-1}$. The bound $f(\alpha) \leq 149$ forces $e = 1$ and $f(\alpha) = 107$.

3 Proof of the fixed-character gap

We prove Lemma 2. Throughout this section $N = C_7$ and $|E| = 243$. Since $[N, E] = N$ and $E' = Z$, we have

$$K' = N \times Z, \quad K/K' \cong E/Z \cong C_3^4.$$

There are 81 linear characters and $149 - 81 = 68$ nonlinear irreducible characters of K .

Suppose first that α acts nontrivially on K/K' . Its dual linear map has a proper fixed subspace, of dimension at most 3. It therefore fixes at most 27 linear characters, and

$$f(\alpha) \leq 27 + 68 = 95. \quad (7)$$

This part does not require the hypothesis on α^3 .

Suppose now that α induces the identity on K/K' . The normal Sylow 7-subgroup N is unique and hence characteristic. The induced automorphism β of $K/N \cong E$ acts trivially on E/Z . Write $\beta(x) = xz_x$ with $z_x \in Z$. Then

$$[\beta(x), \beta(y)] = [x, y],$$

so β fixes $E' = Z$ pointwise. Thus $xZ \mapsto z_x$ is a well-defined linear map $V \rightarrow Z$. The nondegenerate commutator form identifies V with $\text{Hom}(V, Z)$; every such map is given by commutation with an element of E . It follows that β is inner. This argument applies to both extraspecial exponents.

Compose α with a suitable inner automorphism from E so that its induced action on K/N becomes the identity. Its image of E is a Sylow 3-subgroup. Sylow conjugacy and $K = NE$ show that every Sylow 3-subgroup is an N -conjugate of E , since E normalizes itself. A further inner composition from N therefore makes α preserve E , without changing

its action on K/N . The restriction to E is then the identity, because $E \rightarrow K/N$ is an isomorphism.

Consequently, modulo inner automorphisms, α is represented by

$$\gamma_u(a) = a^u \quad (a \in N), \quad \gamma_u(x) = x \quad (x \in E), \quad u \in (\mathbb{Z}/7\mathbb{Z})^\times.$$

These maps are automorphisms: scalar automorphisms of C_7 commute with the given action of E . Inner automorphisms fix every character, so the replacement preserves $f(\alpha)$. It also preserves the condition that the cube is inner, since $\text{Inn}(K)$ is normal in $\text{Aut}(K)$.

Identify $\rho(E)$ with $D = \{1, 2, 4\}$ in the units modulo 7. Every inner automorphism restricts on N to an element of D . Thus $\gamma_u^3 \in \text{Inn}(K)$ implies $u^3 \in D$. Cubes of units modulo 7 are 1 or -1 , and $-1 \notin D$. Therefore $u \in \{1, 2, 4\}$. The case $u = 1$ fixes all 149 characters.

It remains to count the fixed characters for $u = 2$ or 4 . All $k(E) = 83$ irreducibles inflated from K/N are fixed. The other characters lie above the two nonprincipal E -orbits in $\text{Irr}(N)$, each parametrized by $\text{Irr}(T)$ as in Lemma 3, with $k(T) = 33$. Choose $t \in E$ whose action on N is the scalar u , and set $c_t(x) = txt^{-1}$. The automorphism $\delta = c_t^{-1}\gamma_u$ fixes N pointwise and acts on T by $x \mapsto t^{-1}xt$. It has the same action on $\text{Irr}(K)$ as γ_u . In the parametrization above a representative λ , its action on $\lambda \otimes \theta$ is exactly the conjugation action on θ , with no additional linear-character factor. Induction commutes with this action. The fixed contribution above each orbit is therefore

$$h = |\text{Irr}(T)^E|.$$

For completeness, every E -invariant irreducible of T extends to E , since E/T is cyclic of order 3. To see this also when the extension is nonsplit, let τ afford such a character and let A intertwine τ with its conjugate by a lift t of a quotient generator. The operator $A^3\tau(t^3)^{-1}$ is scalar by Schur's lemma. Moreover, A commutes with $\tau(t^3)$. Rescaling A over $\mathbb{C}$ makes $A^3 = \tau(t^3)$ and defines the required extension. The three scalar choices, or twists by the three characters of E/T , give its three distinct extensions. Each noninvariant orbit in $\text{Irr}(T)$ instead induces one irreducible of E . Consequently

$$83 = \frac{33 - h}{3} + 3h, \quad h = 27.$$

It follows that

$$f(\gamma_2) = f(\gamma_4) = 83 + 2 \cdot 27 = 137. \quad (8)$$

Together with (7) and the case $u = 1$, this proves Lemma 2.

Proof of Theorem 1. Section 2 reduces any solution of (2) to $|N| = 7$, $|E| = 243$, $e = 1$ and $f(\alpha) = 107$. Lemma 2 excludes this value. For an extension $K \trianglelefteq H$ with $H/K \cong C_3$, a generator lift t satisfies $t^3 \in K$, so its conjugation automorphism has cube inner. The argument therefore applies to (1) without requiring an order-three lift. $\square$

4 Exact checks, reproducibility and scope

The proof above does not depend on exhaustive computation. Exact checks explain its relation to five concrete kernels: `SmallGroup(1701, i)`, $304 \leq i \leq 308$. Using GAP 4.15.1 and `SmallGrp 1.5.4`, each was checked to have the normal C_7 , an extraspecial Sylow 3-subgroup, and the action specified in Theorem 1. Each has 149 conjugacy classes, with the following distribution:

Class size	1	3	9	21
Number of classes	3	44	48	54

This statement concerns those five identifiers; no completeness claim about the historical search that produced them is needed.

A separate Python implementation used integer coordinates, with no GAP or character-table input. For $v, w \in \mathbb{F}_3^4$, represented by coordinates in $\{0, 1, 2\}$, set

$$c_\varepsilon(v, w) = v_1 w_2 + v_3 w_4 + \varepsilon \left\lfloor \frac{v_1 + w_1}{3} \right\rfloor \pmod{3}, \quad \varepsilon \in \{0, 1\}.$$

The product $(v, z)(w, t) = (v + w, z + t + c_\varepsilon(v, w))$ gives extraspecial models of order 243 and exponents 3 and 9. For each of the 80 nonzero linear forms $\ell : \mathbb{F}_3^4 \rightarrow \mathbb{F}_3$, the corresponding action on additive C_7 is multiplication by $2^{\ell(v)}$. The resulting multiplication on K is

$$(a, v, z)(b, w, t) = (a + 2^{\ell(v)}b, v + w, z + t + c_\varepsilon(v, w)).$$

The computation checked all $2 \cdot 81^3 = 1,062,882$ cocycle triples, all 160 group/action models, and the six scalar maps $(a, v, z) \mapsto (ua, v, z)$ in each model. Conjugation by generators gave complete class partitions. For $u = 1, \dots, 6$ the fixed-class counts were always

$$(149, 137, 83, 137, 83, 83).$$

Fixed-class and fixed-character counts agree because the invertible character table intertwines the two permutation actions, which thus have equal traces. These 960 scalar checks do not enumerate all automorphisms; the normal-form argument supplies that part of the proof. The accompanying reproducibility instructions identify the programs and their exact output records.

The exclusion is restricted to the stated fixed-point-free action with image C_3 and an abelian normal factor of odd order prime to 3. It does not treat other kernels, actions with nontrivial fixed points, or larger action images. In particular, the existence of an odd-order group with commuting probability $1/97$ remains unresolved by these results.

Computational and AI assistance. OpenAI Codex assisted with the proof development, exact GAP and Python checks, mathematical review in a separate model context, and manuscript preparation. The separate review was an automated mathematical audit, not external peer review. The character-theoretic proof is supplied in full; finite computations are corroborating checks and do not certify its infinite quantifiers. No formal proof-assistant verification is claimed.

References

- [1] Basile Beyer de Ryke. The solution to Kourovka problem 21.88. arXiv:2608.03003v1, 2026. <https://arxiv.org/abs/2608.03003v1>.
- [2] Zijian Zeng. Two-generation of the equal-sign fiber product of two symmetric groups, September 2026. Preprint, <https://doi.org/10.5281/zenodo.22293146>.